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martes, 22 de setiembre de 2026 19:09

Cinemática Directa [Scara 3gdl (RRP) - G1-171SZ]

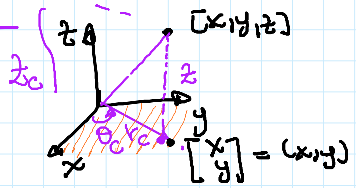
Gdl

Variables Articulares

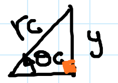
$\begin{bmatrix} J_1 \\ J_2 \\ J_3 \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ d_3 \end{bmatrix}$ → **Cinemática Directa** f(no lineal) → Localización $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ (Posición)

$\begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_6 \end{bmatrix}$ } Kuka

Previo



Cartesianas ⟷ Cilíndricas



Polar:

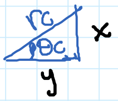
$r_c = \sqrt{x^2+y^2}$

$\theta_c = \tan^{-1}(y|x)$

$r_c = \sqrt{x^2+y^2}$

$\theta_c = \tan^{-1}(y|x)$

$z_c = z$

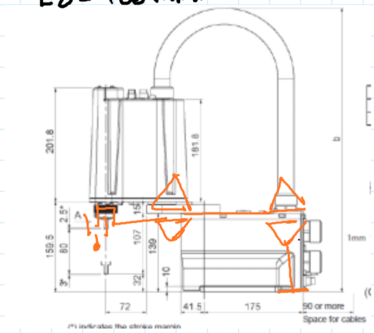


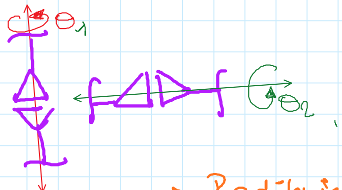
$\left.\begin{aligned} x &= r_c \operatorname{Sin}\theta_c \\ y &= r_c \operatorname{Cos}\theta_c. \end{aligned}\right.$

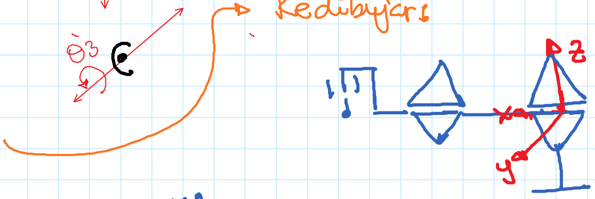
G1-171SZ

L1 = 75mm.

L2 = 100 mm.







$z = -d_3$

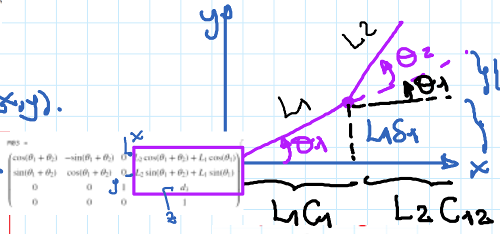
$[\theta_1 y \theta_2] \longrightarrow$ Plano (x,y).

$x = L_1 C_1 + L_2 C_{12}.$

$y = L_1 S_1 + L_2 S_{12}.$

$z = -d_3.$

Función No lineal.



MBS =

$$\begin{pmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) & 0 & L_2\cos(\theta_1+\theta_2)+L_1\cos(\theta_1) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) & 0 & L_2\sin(\theta_1+\theta_2)+L_1\sin(\theta_1) \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

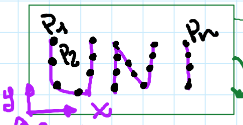
$C_1 = \operatorname{Cos}(\theta_1)$

$S_{12} = \operatorname{Sen}(\theta_1+\theta_2)$

L1 = 75mm.

L2 = 100 mm.

Visión General (Scara G1-171SZ)



Cinemática Inversa → Área de trabajo. } Rango de articulación.

z=-10	θ_1	θ_2	d_3
P1	θ_{11}	θ_{21}	$d_{31} = -10$
P2	θ_{12}	θ_{22}	$d_{32} = -10$
⋮	⋮	⋮	⋮
Pn	θ_{1n}	θ_{2n}	$d_{3n} = -10$

PID - no lineal

$\tau = K_p\tilde{q} + K_v\dot{\tilde{q}} + K_i\varepsilon$

$\dot{\varepsilon} = \tilde{q}.$

; Ec Dinámica Robot

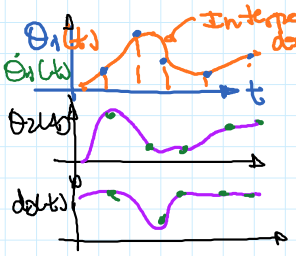
$\tau = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q)$

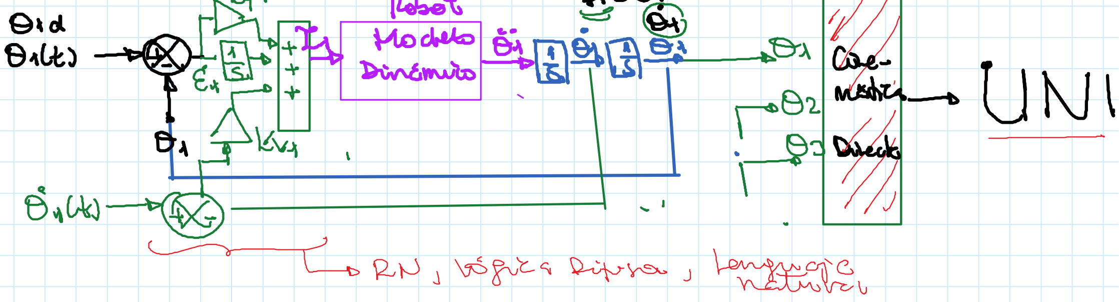
$\tau_1 = M\ddot{\theta}_1 + C\dot{\theta}_1 + G(\theta_1)$

Art 1: $\tau_1 = K_{p1}\tilde{q}_1 + K_{v1}\dot{\tilde{q}}_1 + K_{i1}\varepsilon_1$; $\tilde{q}_1 = \theta_{1d} - \theta_1$

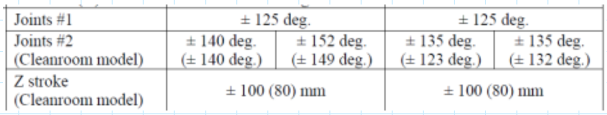
$\dot{\varepsilon}_1 = \tilde{q}_1$

$\dot{\tilde{q}}_1 = \dot{\theta}_{1d} - \dot{\theta}_1$





→ RN, Lógica Difusa, Lenguaje natural



Joints #1	± 125 deg.		± 125 deg.	
Joints #2 (Cleanroom model)	± 140 deg. (± 140 deg.)	± 152 deg. (± 149 deg.)	± 135 deg. (± 123 deg.)	± 135 deg. (± 132 deg.)
Z stroke (Cleanroom model)	± 100 (80) mm		± 100 (80) mm	

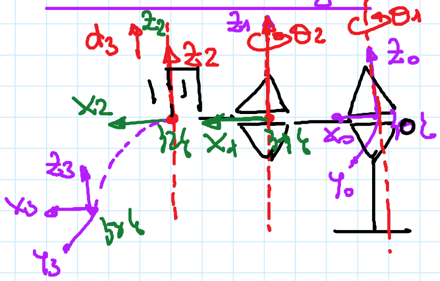
$-125° \leq \theta_1 \leq 125°$

$-135° \leq \theta_2 \leq 135°$

$-100 \leq d_3 \leq 0 \longrightarrow d_3 = -10 = z.$

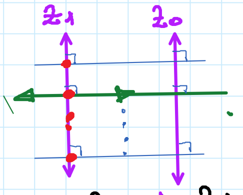
$-135° \leq \theta_2 \leq 135°$
$-100 \leq d_3 \leq 0 \rightarrow d_3 = -10 = z.$

Denavit Hartenberg.

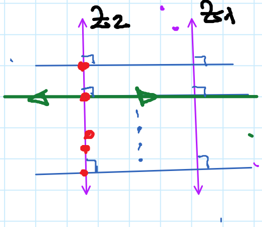


Si $z_i \cap \perp(z_i, z_{i-1})$

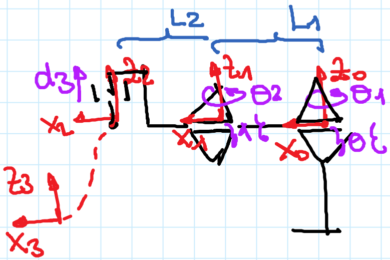
{1} S1 $z_1 \cap \underbrace{\perp(z_1, z_0)}$



$z_2 \cap \underbrace{\perp(z_2, z_1)}$



Algo. DH → Rotz(θ), Tz(d), Tx(a), Rotx(α).



	Rotz θ	Tz d	Tx a	Rotx α
[R] 0-1	θ_1	0	L_1	0°
[R] 1-2	θ_2	0	L_2	0°
[P] 2-3	0°	d_3	0	0

$\theta_1 = \theta_2 = d_3 = 0$